# Matplotlib Data Visualization Workflow

Data bujhte hole chart dekha dorkar. Eta Matplotlib shikhar step-by-step guide, jeta Pandas-er sathe prai sob dataset-e kaj lage. Protita method er pase comment diye bojhano hoyeche.

## Workflow ek nojore

**Setup → Prothom Chart → Customize → Chart Type Bachai → Subplots → Pandas theke Plot → Annotate → Style → Save**

Ei order follow korle jekono dataset-er chart banano shohoj hoye jay.

### Guide-er example data

Ei guide-er Pandas-er example-e `sales.csv` file use hoyeche, jeikhane column gulo: `date, city, price, qty, total`. Apni nijer dataset-er column-er nam boshiye nien.

## 1. Setup ar Import

```
import matplotlib.pyplot as plt   # chart banar main module (sobsomoy plt nam e)
import pandas as pd                # data niye kaj korar jonno
import numpy as np                 # number/array er jonno

# Jupyter Notebook e chart notebook er moddhei dekhate hole:
# %matplotlib inline
```

### Matplotlib-er duto jinis mone rakhte hobe:

| Concept / Method | Mane |
|---|---|
| `Figure (fig)` | Pura canvas ba chhobi. Eta-r moddhe ek ba ekadhik chart thake |
| `Axes (ax)` | Figure-er moddhe ekta chart. X axis, Y axis, title shob ei ax-er |
| `plt.subplots()` | Ekshathe fig ar ax banay. Prai shobsomoy eta diye shuru korben |
| `plt.show()` | Chart ta screen e dekhay. Chart ankar por shobar shesh e likhte hoy |

Tip: Matplotlib-e duto style ache. **plt.plot()** (shohoj, shudhu ekta chart-er jonno) ar **fig, ax = plt.subplots()** (professional, ekadhik chart-er jonno). Ekhon theke dwitiyo-ta obhyash korun.

## 2. Prothom Chart

### Shohoj style (plt):

```
months = [1, 2, 3, 4, 5, 6]
sales = [120, 150, 130, 180, 210, 240]

plt.plot(months, sales)            # line chart ankhe
plt.title("Monthly Sales")         # chart er nam
plt.xlabel("Month")                # x axis er nam
plt.ylabel("Sales (taka)")         # y axis er nam
plt.show()                         # chart ta screen e dekhay
```

### Recommended style (fig, ax):

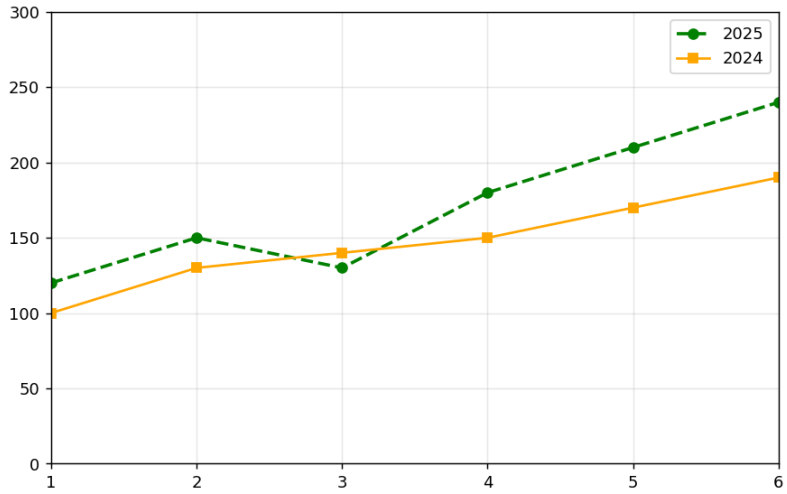
```
fig, ax = plt.subplots(figsize=(8, 5))   # fig = canvas, ax = chart (8 x 5 inch)
ax.plot(months, sales)                   # chart ankar kaj ax diye hoy
ax.set_title("Monthly Sales")            # ax e set_ diye likhte hoy
ax.set_xlabel("Month")
ax.set_ylabel("Sales (taka)")
plt.show()
```

| Method | Ki kaj kore |
|---|---|
| `plot(x, y)` | X ar Y value diye line chart ankhe |
| `title / set_title` | Chart-er nam dey |
| `xlabel / set_xlabel` | X axis-er nam dey |
| `ylabel / set_ylabel` | Y axis-er nam dey |
| `figsize=(w, h)` | Chart-er size inch-e (width, height) |

## 3. Customize (chart shundor kora)

```
sales_2024 = [100, 130, 140, 150, 170, 190]

fig, ax = plt.subplots(figsize=(8, 5))
ax.plot(months, sales, color="green", linestyle="--", marker="o",
        linewidth=2, label="2025")          # label na dile legend khali thakbe
ax.plot(months, sales_2024, color="orange", marker="s", label="2024")
ax.legend()                                 # label gulo box e dekhay
ax.grid(True, alpha=0.3)                    # halka grid line
ax.set_xlim(1, 6)                           # x axis er range
ax.set_ylim(0, 300)                         # y axis er range
plt.show()
```



| Method | Ki kaj kore |
|---|---|
| `color` | Rong. Nam ("red") ba hex code ("#2E7D6B") dewa jay |
| `linestyle` | "-" (normal), "--" (dash), ":" (dot) |
| `marker` | Data point-er chinho. "o" (gol), "s" (chokor), "^" (tikon) |
| `linewidth` | Line-er motha |
| `label + legend()` | Protita line-er nam; legend() sheta box e dekhay |
| `grid()` | Pichone grid line dey, pora shohoj hoy |
| `set_xlim / set_ylim` | Axis-er range thik kore dey |
| `alpha` | Shocchho-bhab (0 = ekdom shocchho, 1 = purno rong) |

# 4. Chart Type Bachai

Sobcheye boro bhul: bhul chart bachai kora. Age bhebe nin apni ki dekhate chan:

| Chart | Kokhon use korbe |
|---|---|
| `plot() (line)` | Somoy-er sathe poriborton (trend). Jemon mash onujayi bikri |
| `bar() / barh()` | Category gulo tulona. Jemon city wise sales |
| `hist()` | Ekta number-er spread (distribution). Jemon price kothay beshi |
| `scatter()` | Duto number-er sombondho. Jemon price vs qty |
| `boxplot()` | Median, quartile ar outlier ek nojore |
| `pie()` | Bhager obosta. Shudhu 5 tar kom part thakle use korun |

## Bar chart

```
cities = ["Dhaka", "Chattogram", "Khulna", "Sylhet"]
totals = [500, 320, 180, 150]

fig, ax = plt.subplots()
ax.bar(cities, totals, color="steelblue")   # ulombo (vertical) bar
# ax.barh(cities, totals)                   # shoyano (horizontal) bar
ax.set_title("City wise Sales")
plt.show()
```

## Histogram

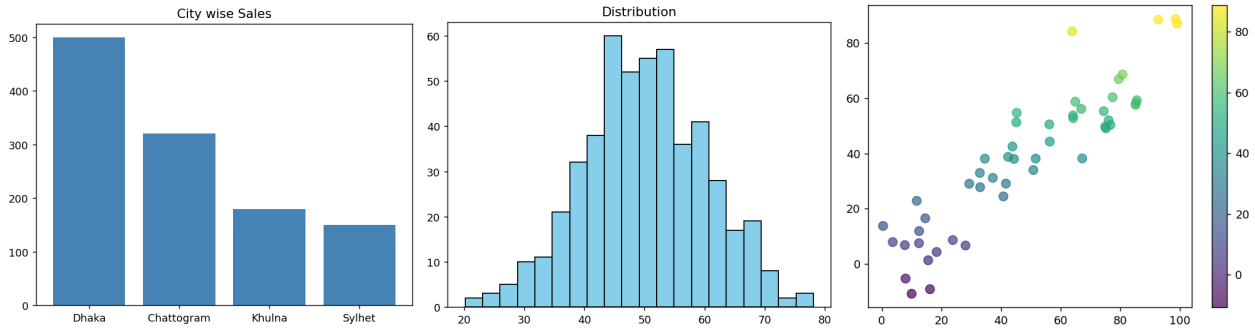
```
data = np.random.normal(50, 10, 500)    # 500 ta random number (average 50)

fig, ax = plt.subplots()
ax.hist(data, bins=20, color="skyblue", edgecolor="black")  # 20 ta bin e bhag
ax.set_title("Distribution")
plt.show()
```

## Scatter plot

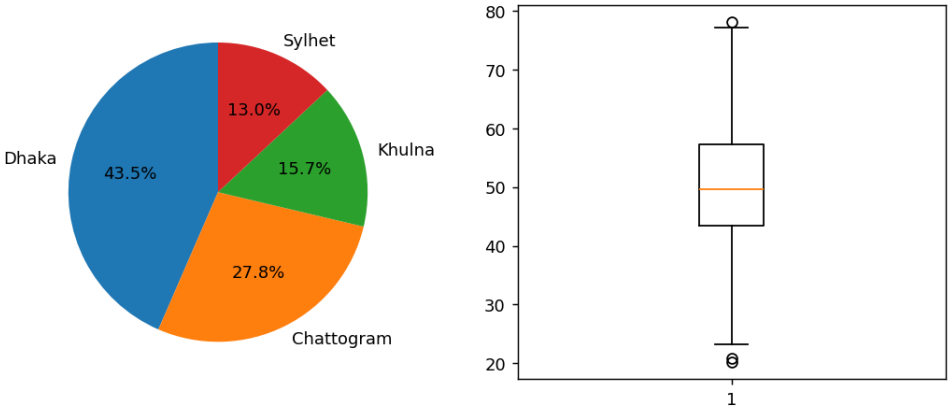
```
x = np.random.rand(50) * 100
y = x * 0.8 + np.random.randn(50) * 10

fig, ax = plt.subplots()
sc = ax.scatter(x, y, c=y, cmap="viridis", s=60, alpha=0.7)
# c=y -> value onujayi rong, s -> dot er size, alpha -> shocchho-bhab
fig.colorbar(sc, ax=ax)                     # rong er scale dekhay
plt.show()
```



### Pie chart ar Box plot

```
fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(10, 4))   # 1 row, 2 chart
ax1.pie(totals, labels=cities, autopct="%1.1f%%", startangle=90)
ax2.boxplot(data)                    # median, quartile, outlier ek nojore
plt.show()
```

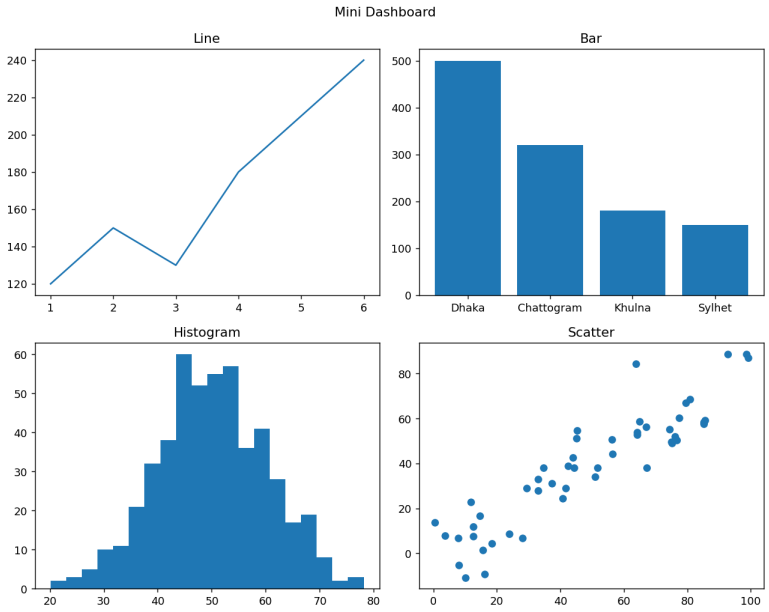



## 5. Subplots (ekta figure-e onek chart)

```
fig, axes = plt.subplots(2, 2, figsize=(10, 8))   # 2 row x 2 column

axes[0, 0].plot(months, sales)              # upore bam
axes[0, 0].set_title("Line")
axes[0, 1].bar(cities, totals)              # upore dan
axes[0, 1].set_title("Bar")
axes[1, 0].hist(data, bins=20)              # niche bam
axes[1, 0].set_title("Histogram")
axes[1, 1].scatter(x, y)                    # niche dan
axes[1, 1].set_title("Scatter")

fig.suptitle("Mini Dashboard")              # pura figure er title
plt.tight_layout()                          # chart gulo jeno gaye gaye na lage
plt.show()
```



| Method | Ki kaj kore |
|---|---|
| `subplots(r, c)` | r ta row, c ta column-er chart-er grid banay |
| `axes[0, 1]` | Grid-er kon ghor, seta bole (row, column). Count 0 theke shuru |
| `suptitle()` | Pura figure-er boro title |
| `tight_layout()` | Chart gulo jeno ekta arekta-r upor na pore, seta thik kore |
| `sharex / sharey` | subplots(1, 2, sharey=True) -> duto chart-er y axis ek-i scale |

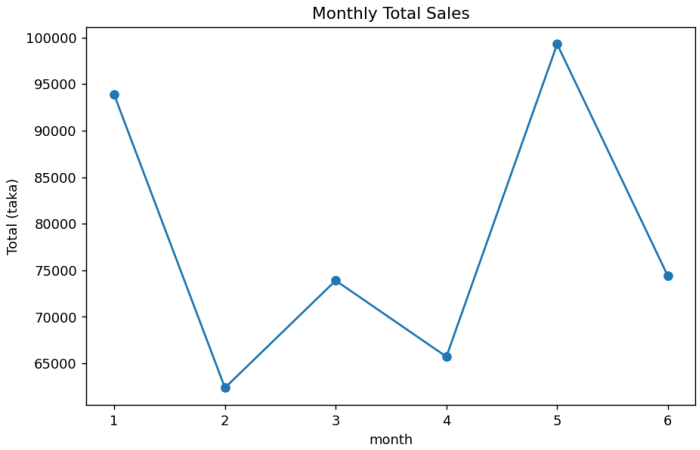
## 6. Pandas theke Plot (sobcheye dorkari)

Real kaje data DataFrame-e thake. Age groupby kore summary banan, tarpor plot korun.

```
df = pd.read_csv("sales.csv")
df["date"] = pd.to_datetime(df["date"])
df["month"] = df["date"].dt.month

monthly = df.groupby("month")["total"].sum()      # mash onujayi total

fig, ax = plt.subplots(figsize=(8, 5))
monthly.plot(kind="line", marker="o", ax=ax)      # ax=ax -> amar ax e anko
ax.set_title("Monthly Total Sales")
ax.set_ylabel("Total (taka)")
plt.show()
```




```
city_sales = df.groupby("city")["total"].sum().sort_values(ascending=False)

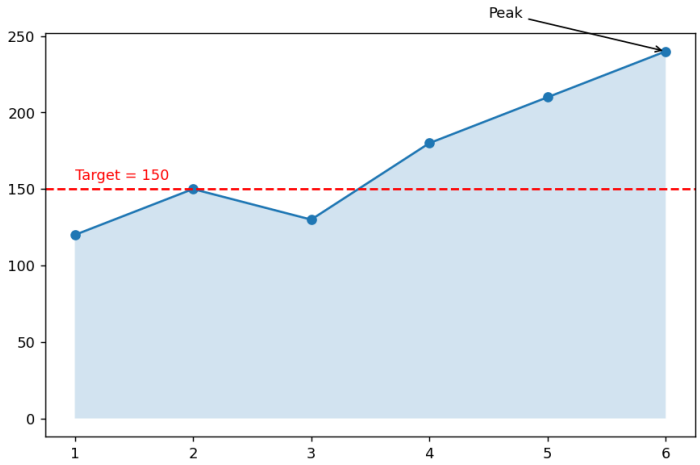
fig, ax = plt.subplots(figsize=(8, 5))
city_sales.plot(kind="bar", color="teal", ax=ax)
ax.tick_params(axis="x", rotation=45)             # x er lekha 45 degree ghurai
plt.tight_layout()
plt.show()

# df["price"].plot(kind="hist", bins=20)          # histogram
# df.plot(kind="scatter", x="price", y="qty")     # scatter
```

| Method | Ki kaj kore |
|---|---|
| `df.plot(kind=...)` | Pandas-er built-in plot, bhitore Matplotlib-i use kore |
| `kind=` | "line", "bar", "barh", "hist", "box", "pie", "scatter" |
| `ax=ax` | Pandas-er plot ke amar nijer ax-e boshay, tai title/label dewa jay |
| `tick_params(rotation=45)` | X axis-er lekha ghurai, lomba nam thakle dorkar |
| `sort_values()` | Bar chart ke boro theke choto sajay, dekhte bhalo lage |

# 7. Annotate (chart-e extra tothyo)

```
fig, ax = plt.subplots(figsize=(8, 5))
ax.plot(months, sales, marker="o")
ax.annotate("Peak", xy=(6, 240), xytext=(4.5, 262),
            arrowprops=dict(arrowstyle="->"))      # tir diye likhe dekhay
ax.axhline(y=150, color="red", linestyle="--")     # ekta ashar (horizontal) line
ax.text(1, 156, "Target = 150", color="red")       # jekono jaygay lekha
ax.fill_between(months, sales, alpha=0.2)          # line er niche rong
plt.show()
```




| Method | Ki kaj kore |
|---|---|
| `annotate()` | Nirdishto point-e tir diye lekha dey. Jemon "Peak" |
| `axhline() / axvline()` | Ashar ba ulombo line tane. Jemon average ba target |
| `text(x, y, "...")` | Jekono jaygay lekha boshay |
| `fill_between()` | Line-er niche rong dey |

# 8. Style ar Colormap

```
print(plt.style.available)       # kon kon style ache dekhte

plt.style.use("ggplot")         # sob chart e ekshathe style lage
# plt.style.use("default")      # abar age er obostha e jete

# colormap: "viridis", "plasma", "coolwarm", "Blues"
# ax.scatter(x, y, c=y, cmap="coolwarm")
```

| Method | Ki kaj kore |
|---|---|
| `style.use("ggplot")` | Ekshathe sob chart-er look bodlay |
| `style.available` | Kon kon style ache tar list |
| `cmap=` | Value onujayi rong-er palette (viridis, coolwarm ityadi) |
| `colorbar()` | Rong-er scale-ta chart-er pashe dekhay |

Style-er jonno ei gulo try korun: `ggplot`, `seaborn-v0_8-whitegrid`, `fivethirtyeight`, `bmh`.

## 9. Save

```
fig, ax = plt.subplots(figsize=(8, 5))
ax.plot(months, sales)

fig.savefig("sales_chart.png", dpi=300, bbox_inches="tight")  # PNG, high quality
fig.savefig("sales_chart.pdf")                                # PDF/SVG o hoy
plt.close(fig)                  # memory khali kore (onek chart banale dorkar)
```

| Method | Ki kaj kore |
|---|---|
| `savefig()` | Chart-ke PNG, PDF ba SVG file hishebe save kore |
| `dpi=300` | Chobir quality. Report ba CV-r jonno 300 bhalo |
| `bbox_inches="tight"` | Chobir kinara theke extra faka kete dey, label kete jay na |
| `plt.close(fig)` | Chart bondho kore memory khali kore |

Shotorkota: **savefig()** sobshomoy **plt.show()**-er age likhun. Jupyter-e kono shomossha hoy na, kintu script-e show()-er pore save korle khali chobi save hote pare.

# Cheat Sheet (choto mone rakhar jonno)

| Kaj | Method |
|---|---|
| `Shuru` | fig, ax = plt.subplots(figsize=(8, 5)) |
| `Line / Bar` | ax.plot() , ax.bar() , ax.barh() |
| `Distribution` | ax.hist() , ax.boxplot() |
| `Relation` | ax.scatter() |
| `Part of whole` | ax.pie() |
| `Nam dewa` | set_title , set_xlabel , set_ylabel , legend |
| `Shajano` | color , linestyle , marker , grid , set_xlim |
| `Onek chart` | plt.subplots(2, 2) , tight_layout |
| `Pandas theke` | df.plot(kind="bar", ax=ax) |
| `Save` | fig.savefig("a.png", dpi=300, bbox_inches="tight") |

Tip: Kono dataset pelei ei order-e kaj shuru koro: **Pandas diye clean + groupby → chart type bachai → ax diye anko → title/label dao → save koro**.

Porer dhap: **Seaborn**. Eta Matplotlib-er upor bana, kom code-e shundor statistical chart (heatmap, pairplot, boxplot) banay. Matplotlib bujhle Seaborn onek shohoj lagbe.